TECHNICAL INFORMATION: AY-3-8914 (0098-0660) U6

AY-3-8914*

PROGRAMMABLE SOUND GENERATOR

FEATURES

- Full software control of sound generation
- Interfaces to most 8-bit and 16-bit microprocessors
- Three independently programmed analog outputs
- Two 8-bit general purpose I/O ports
- Single +5 Volt Supply

DESCRIPTION

The AY-3-8914 Programmable Sound Generator (PSG) is a Large Scale Integrated Circuit which can produce a wide variety of complex sounds under software control. The AY-3-8914 is manufactured in GI's N-Channel Ion Implant Process. Operation requires a single 5V power supply, a TTL compatible clock and a microprocessor controller such as the GI 16-bit CP1600/1610 or one of GI's PIC 1650 series of 8-bit microcomputers.

The PSG is easily interfaced to any bus oriented system. Its flexibility makes it useful in applications such as music synthesis, sound effects generation, audible alarms, tone signalling and FSK modems. The analog sound outputs can each provide 4 bits of logarithmic digital analog conversion, greatly enhancing the dynamic range of the sounds produced.

In order to perform sound effects while allowing the processor to continue its other tasks, the PSG can continue to produce sound after the initial commands have been given by the control processor. The fact that realistic sound production often involves more than one effect is satisfied by the three independently controllable channels available in the PSG.

All of the circuit control signals are digital in nature and intended to be provided directly by a microprocessor/microcomputer. This means that one PSG can produce the full range of required sounds with no change in external circuitry. Since the frequency response of the PSG ranges from subaudible at its lowest frequency to postaudible at its highest frequency, there are few sounds which are beyond reproduction with only the simplest electrical connections.

Since most applications of a microprocessor/PSG system would also require interfacing between the outside world and the microprocessor, this facility has been designed into the PSG. The AY-3-8914 has two general purpose 8-bit I/O ports and is supplied in a 40 lead package.

PIN FUNCTIONS

DA7—DA0 (input/output/high impedance) pins 30-37 (AY-3-8914)
Data/Address 7—0:

These 8 lines comprise the 8-bit bidirectional bus used by the microprocessor to send both data and addresses to the PSG and to receive data from the PSG in the data mode. DA-7—DA0 correspond to Register Array bits B-7—B0 in the address mode. DA3—DA0 select the register # (0—17s) and DA7—DA4 in conjunction with address inputs A9 and A8 form the high order address (chip select)

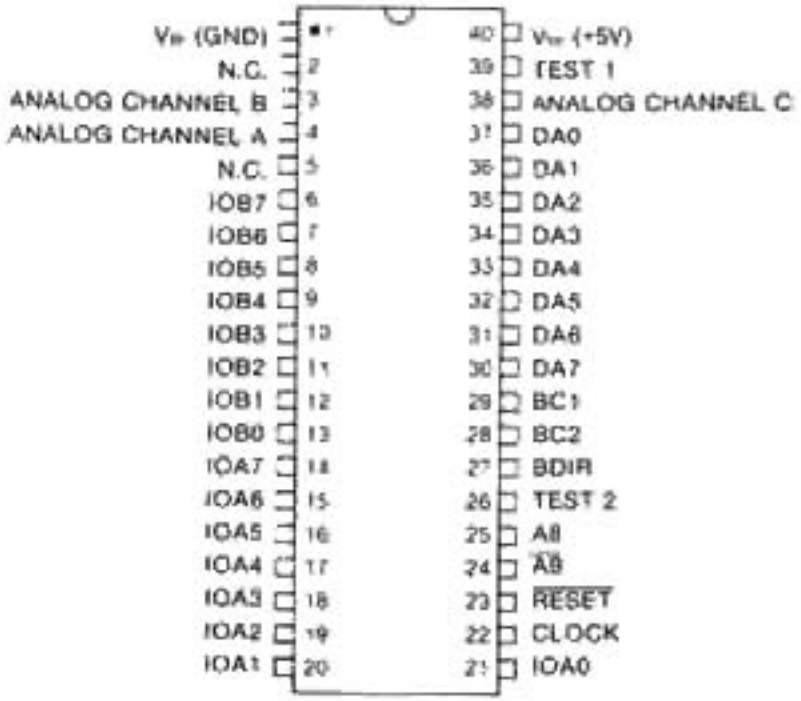
$\overline{A8}$ (input) pin 25 (AY-3-8914)
A9 (input) pin 24 (AY-3-8914)

Address 9. Address 8

These "extra" address bits are made available to enable the positioning of the PSG (assigning a 16 word memory space) in a total 1.024 word memory area rather than in a 256 word memory area as defined by address bits DA7—DA0 alone. If the memory size does not require the use of these extra address lines, they may be left unconnected as each is provided with either an on-chip pull down (A9) or pull-up (A8) resistor in "noisy"environments, however, it is recommended that A9 and A8 be tied to an external ground and +5V respectively if they are not to be used.

Pin Configurations
40 Lead Dual In Line

Top View



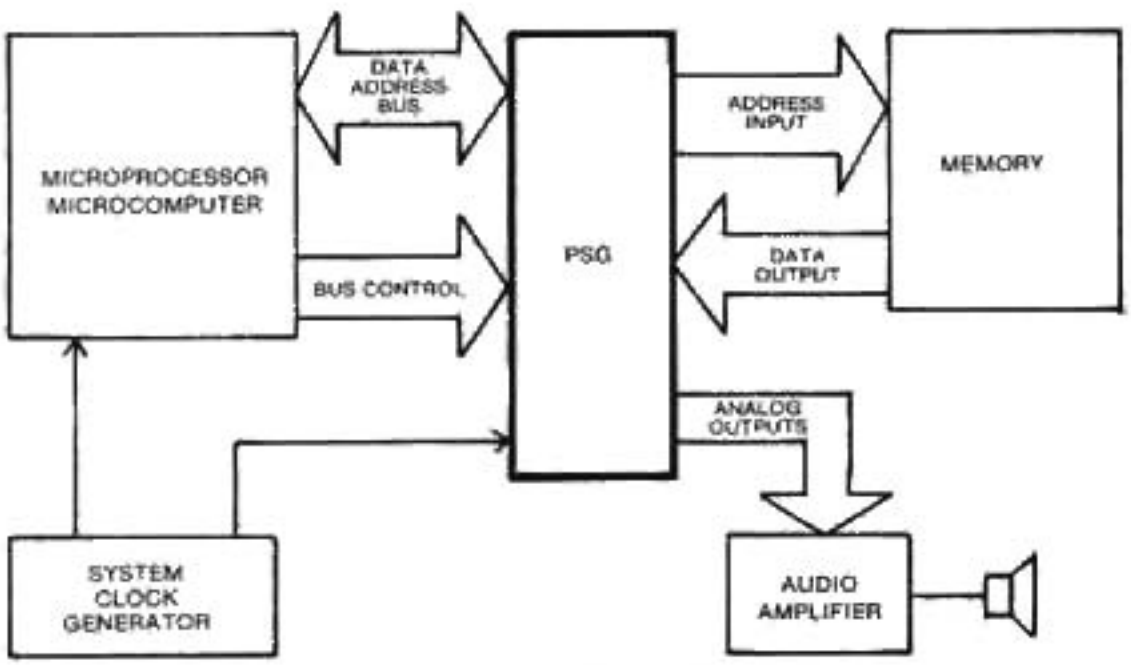


Fig. 1 SYSTEM BLOCK DIAGRAM

*For more detailed data refer to General Instrument data specification book.